

1.1 - Analytic Stability Boundary for a Performative Market Maker

REFLEX project - math-theory

STATUS: DONE (Priority 1). This document discharges methodology target **1.1**. It derives, in closed form, the performative-prediction contraction modulus m of the REFLEX policy–distribution loop directly from the microstructure primitives of the `endo_market_v2` simulator, and obtains the stability boundary

$$\text{stable} \iff m = \frac{\varepsilon \beta}{\gamma} < 1 \iff \varepsilon < \frac{\gamma}{\beta},$$

with γ , β , and ε each given as a closed-form function of the model parameters - none swept, none tuned. The result is then turned into a falsifiable cross-check against the existing model-free estimator in `analysis/response_modulus.py`.

Why this is the headline contribution. Perdomo et al. (ICML 2020) prove that the repeated-retraining map contracts iff $\varepsilon < \gamma/\beta$, but treat γ (strong convexity), β (joint smoothness) and ε (distribution sensitivity) as abstract Lipschitz constants of an unspecified loss. For a structural market-making model these constants are not free: they are pinned by the fill-intensity curve, the adverse-selection channel, and the quoting-cost friction. We compute them. The deliverable is an *a-priori* stability boundary: predict whether a given bond/dealer configuration converges before running a single RRM loop, then verify the prediction against the model-free best-response (BR) slope probe.

1 Notation

Symbol	Reading	Definition / source
h	central half-spread (the policy coordinate)	bias b_h of <code>LinearPolicy</code>
h^*	self-consistent fixed-point half-spread	root of the FOC, §4
h_{dep}	deployed half-spread setting the toxic env.	§1, A3
$J(h; T)$	expected one-step dealer objective	§2
$\tau(h)$	expected informed (toxic) notional	§2
T	frozen toxic level in one deployment, $T = \tau(h_{\text{dep}})$	§1, A3
ε	distribution sensitivity, $= d\tau/dh $	§3.1
β	joint smoothness, $= \partial^2 J / \partial h \partial T $	§3.2
γ	strong convexity, $= -\partial^2 J / \partial h^2$	§3.3
m	contraction modulus $= \text{BR}'(h^*) = \varepsilon\beta/\gamma$	§3.4
ρ	liquidity ratio <code>liquidity/liq_mean</code>	A2
g, u	mispricing $v - m$, standardized $u = g/\sigma_s$	§2
$\bar{g}(u)$	gate mean (a 1-D Gaussian integral)	§2

2 Setup and assumptions

We analyse the loop at the level of the **central half-spread** h - the bias coordinate b_h of the dealer policy, which the convergence study identifies as the dominant coordinate of the policy-to-distribution iterate map. The model-free probe `response_modulus.py` deploys constant- h , zero-skew policies; we adopt the same restriction so the analytic and empirical moduli measure the same object.

Assumption 1 (single representative bond, zero skew). *One bond, skew = 0. The cross-sectional factor structure rescales but does not change the scalar boundary; the multi-bond lift is Priority 1.5.*

Assumption 2 (reference state). *Curvature is evaluated at the probe’s reference state $s_0 = \text{simulator.reset}$ with inventory $q_0 \approx 0$, mispricing $g = v - m$, and liquidity ratio $\rho = \text{liquidity}/\text{liq_mean}$. Expectations below are over the client-flow noise at that state.*

Assumption 3 (frozen-environment best response). *This is the structural content of “the operator is blind to $dD/d\phi$ ”. Within a single deployment the learned operator T_θ conditions on a frozen policy summary of the deployed regime. So when the dealer computes its best response it sees the toxic-flow environment as fixed at the level induced by the deployed spread h_{dep} (write $T = \tau(h_{\text{dep}})$), while the benign demand-elasticity channel still responds to its candidate h . The performative feedback enters only across deployments - exactly the cobweb the loop iterates.*

Assumption 4 (saturation non-binding). *The informed-flow cap $\text{info_cap} \cdot \tanh(\cdot/\text{info_cap})$ is in its linear regime at the operating point. We carry the uncapped form and flag the saturated regime as a documented corollary (§5.3).*

3 The expected single-step objective $J(h)$

3.1 Components from the simulator

At zero skew, with S = dealer sells (clients lift the ask), B = dealer buys (clients hit the bid), $q_{\text{after}} = q_0 + B - S$, v = fundamental, v' = next fundamental, m = mid:

$$\begin{aligned}\text{spread_capture} &= (S + B) h, \\ \text{inventory_pnl} &= q_{\text{after}} (v' - v), \\ \text{adverse_selection_loss} &= (B - S) (m - v).\end{aligned}$$

The dealer objective is

$$J = P [\text{spread_capture} + \text{inventory_pnl} - \text{adverse_selection_loss} - \text{inv_risk_weight} \cdot q_{\text{after}}^2 - w (h - 2)],$$

with $w = \text{quote_anchor_weight}$, quote_anchor_ref , and $P := \text{pnl_scale}$.

3.2 Uninformed (benign) flow

Benign notional has the Guéant–Lehalle–Fernández-Tapia / Avellaneda–Stoikov exponential fill intensity

$$U(h) = A e^{-kh} \rho, \quad A = \text{base_arrival_rate}, \quad k = \text{demand_elasticity},$$

split roughly 50/50 into buys/sells with a zero-mean idiosyncratic imbalance, so $\mathbb{E}[S_{\text{uninf}}] = \mathbb{E}[B_{\text{uninf}}] = \frac{1}{2}U(h)$ and $\mathbb{E}[S_{\text{uninf}} - B_{\text{uninf}}] = 0$.

3.3 Informed (toxic) flow

With $\text{signal} = g + \sigma_s \eta$, $\eta \sim N(0, 1)$, $\sigma_s = \text{info_signal_noise}$, the gate is $\text{gate} = \tanh(|\text{signal}|/\sigma_s)$. The informed notional is

$$\tau(h) = \rho \bar{g} (I_b + \alpha f I e^{-c_t h}),$$

with $I_b = \text{info_base_intensity}$, $I = \text{info_intensity}$, $\alpha = \text{alpha}$, $f = \text{toxicity_feedback}$, $c_t = \text{info_spread_decay}$, and $\bar{g} = \mathbb{E}_\eta[\tanh(|g + \sigma_s \eta|/\sigma_s)]$ the gate mean. Informed traders pick the dealer off, so $\mathbb{E}[(B - S)_{\text{inf}}] = -\tau(h) \text{sign}(g)$ and the expected adverse-selection loss is

$$\mathbb{E}[\text{adverse}] = \psi \tau(h), \quad \psi > 0.$$

3.4 Inventory terms vanish or are second order

The fundamental increment $v' - v = \sigma_v$ shock is drawn after, and independently of, q_{after} , so $\mathbb{E}[\text{inventory_pnl}] = \mathbb{E}[q_{\text{after}}] \cdot 0 = 0$ for every h . The quadratic risk $\mathbb{E}[q_{\text{after}}^2]$ is $O(q_0^2)$ plus a flow-variance term that is second order at $q_0 \approx 0$; we carry it as a small additive curvature $\lambda_q \geq 0$ that only raises γ (it is stabilising) and never changes the boundary's form.

3.5 The two Gaussian integrals

Standardize $u = g/\sigma_s$. Because $\tanh$ is odd, $\tanh(|x|) \text{sign}(x) = \tanh(x)$, so the two state constants reduce to clean 1-D Gaussian integrals in u alone:

$$\begin{aligned} \bar{g}(u) &= \mathbb{E}_{\eta \sim N(0,1)}[\tanh(|u + \eta|)], \\ a(u) &= \mathbb{E}_{\eta \sim N(0,1)}[\tanh(u + \eta)], \\ \psi &= \sigma_s u \frac{a(u)}{\bar{g}(u)} > 0. \end{aligned}$$

Both are smooth, bounded, monotone-in- $|u|$ integrals evaluated by quadrature. They are state constants (independent of h), so they pass through every h -derivative untouched.

3.6 Assembled objective

Writing the toxic level as the frozen $T = \tau(h_{\text{dep}})$ (A3):

$$J(h; T) = P \left[\underbrace{h A e^{-kh} \rho}_{\text{uninf. spread}} + \underbrace{h T}_{\text{toxic spread}} - \underbrace{\psi T}_{\text{adverse sel.}} - \underbrace{w(h - 2 - \lambda_q)}_{\text{quoting cost}} \right]. \quad (1)$$

Key structural fact. The adverse term $-\psi T$ is constant in h (the toxic level is frozen within a deployment), so it drops out of every h -derivative below. It sets the level of profit - the echo-chamber gap of Priority 1.2 - but *adverse selection does not enter the stability boundary*, only the optimal-vs-stable profit gap.

4 The best-response map and its slope

4.1 The map

The dealer best-responds by maximising $J(h; T)$ at the frozen toxic level $T = \tau(h_{\text{dep}})$: $\text{BR}(h_{\text{dep}}) = \arg \max_h J(h; \tau(h_{\text{dep}}))$. The RRM iteration is $h_{k+1} = \text{BR}(h_k)$; linearising at a fixed point h^* , $h_{k+1} - h^* \approx \text{BR}'(h^*)(h_k - h^*)$, so the loop contracts iff the modulus $m := |\text{BR}'(h^*)| < 1$.

4.2 First-order condition and the three derivatives

Define $F(h, T) := \partial J / \partial h$:

$$F(h, T) = P [A\rho e^{-kh}(1 - kh) + T - 2w(h -)], \quad F = 0 \text{ at the optimum.} \quad (2)$$

(i) Curvature in h (gives γ). Only the uninformed term and quoting cost depend on h (hT is linear):

$$\begin{aligned} \frac{d}{dh} [A\rho e^{-kh}(1 - kh)] &= A\rho e^{-kh} [-k(1 - kh) - k] = A\rho e^{-kh} (k^2 h - 2k) \\ &= A\rho k e^{-kh} (kh - 2), \\ \frac{d}{dh} [-2w(h -)] &= -2w, \end{aligned}$$

hence $\partial^2 J / \partial h^2 = P [A\rho k e^{-kh} (kh - 2) - 2w]$ and

$$\boxed{\gamma := -\frac{\partial^2 J}{\partial h^2} = P [2w + A\rho k e^{-kh} (2 - kh) + \lambda_q].} \quad (3)$$

(ii) Cross sensitivity (gives β). Only hT carries a T -dependence in the gradient (coefficient +1); the constant adverse term contributes nothing to $\partial J / \partial h$:

$$\boxed{\beta := \left| \frac{\partial^2 J}{\partial h \partial T} \right| = P = \text{pnl_scale}.} \quad (4)$$

(iii) Distribution sensitivity (gives ε).

$$\boxed{\varepsilon := \left| \frac{d\tau}{dh} \right| = \rho \bar{g} \alpha f I c_t e^{-c_t h}.} \quad (5)$$

ε is the slope of the toxic-flow response to the deployed spread (tighter quotes summon more toxic flow) and is linear in α and in `toxicity_feedback` = f .

4.3 Implicit-function theorem: the BR slope

Differentiate $F(h^*(h_{\text{dep}}), \tau(h_{\text{dep}})) = 0$ totally in h_{dep} , with $F_h = -\gamma$, $F_T = +\beta$, $d\tau/dh_{\text{dep}} = -\varepsilon$:

$$\frac{dh^*}{dh_{\text{dep}}} = -\frac{F_T (d\tau/dh_{\text{dep}})}{F_h} = -\frac{(\beta)(-\varepsilon)}{(-\gamma)} = -\frac{\beta \varepsilon}{\gamma}.$$

So $\text{BR}'(h_{\text{dep}}) = -\beta \varepsilon / \gamma$ - a *negative* slope. The cobweb is oscillatory: deploying wider invites a narrower best response and vice versa.

4.4 The modulus and the boundary

Theorem 1 (analytic stability boundary). *At the fixed point h^* ,*

$$m(h^*) = |\text{BR}'(h^*)| = \frac{\varepsilon \beta}{\gamma} = \frac{\rho \bar{g} \alpha f I c_t e^{-c_t h^*}}{2w + A\rho k e^{-kh^*} (2 - kh^*) + \lambda_q}, \quad (6)$$

(the common $P = \text{pnl_scale}$ cancels), and the loop is stable iff

$$\varepsilon < \frac{\gamma}{\beta} \iff \rho \bar{g} \alpha f I c_t e^{-c_t h^*} < 2w + A\rho k e^{-kh^*} (2 - kh^*) + \lambda_q.$$

Every quantity is a closed form in the config and h^* , so the inequality is an a-priori predictor. $m = |\text{BR}'|$ is exactly what `response_modulus.py` returns, so (6) is the directly comparable prediction (§7).

5 Locating the fixed point h^*

The boundary is evaluated where the deployed spread is its own best response. Setting $F(h, \tau(h)) = 0$ with the toxic level evaluated self-consistently:

$$A\rho e^{-kh^*}(1 - kh^*) + \tau(h^*) - 2w(h^* - = 0, \quad \tau(h^*) = \rho\bar{g}(I_b + \alpha f I e^{-c_t h^*}). \quad (7)$$

A smooth scalar equation; solve by Newton or bisection on $[0, \text{max_half_spread}]$. A unique interior root exists wherever $\gamma(h) > 0$ (strict concavity of J in h) - exactly the stability condition holding with room to spare.

6 Corollaries (each a falsifiable prediction)

Corollary 1 (linear-in-adversariality scaling). *At fixed h^* , m is linear in αf , and $m = 1$ at*

$$(\alpha f)^* = \frac{\gamma}{\rho \bar{g} I c_t e^{-c_t h^*}},$$

predicting the `toxicity_feedback` value at which `sweep_feedback.yaml` crosses the boundary before the sweep is run.

Corollary 2 (lazy-deploy raises effective curvature). *Under `update_rule="rgd"` with step η , one gradient step gives the map $M(h) = h + \eta G(h)$ with $G(h) = \partial_h J(h; \tau(h))$ and linearised modulus $m_{\text{RGD}} = |1 - \eta(\gamma - \varepsilon\beta)|$ in the worst-case (destabilising) orientation. Contraction requires $\gamma_{\text{eff}} := \gamma - \varepsilon\beta > 0$, i.e. the same boundary $\varepsilon < \gamma/\beta$. Taking K lazy steps compounds the contraction, enlarging the stable region.*

Corollary 3 (the modulus saturates rather than blows up). *As h^* grows, $A\rho k e^{-kh^*}(2 - kh^*) \rightarrow 0$, so $\gamma \rightarrow 2wP$ (a floor, not zero) while ε also decays. Their ratio approaches a finite plateau $m_\infty \approx \varepsilon(h^*)/(2w)$ rather than diverging - the analytic explanation of the empirically observed “modulus saturates (~ 1.25) past the boundary”. The GLFT term even turns negative for $kh^* > 2$ (defensive widening, low curvature).*

Corollary 4 (why ε , not α , is the clean control variable). *Sweeping α moves both ε (numerator, $\propto \alpha$) and h^* (the dealer widens), feeding back through $e^{-c_t h^*}$ and the γ curvature term, so $m(\alpha)$ is confounded and its sign is not robust to universe size. Sweeping $f = \text{toxicity_feedback}$ at fixed structural regime moves ε linearly with h^* far less perturbed, isolating the performative channel.*

7 Symbol $\rightarrow$ config map

Symbol	Meaning	Config field	Default
A	uninformed arrival scale	<code>clients.base_arrival_rate</code>	1.0
k	demand elasticity (GLFT decay)	<code>clients.demand_elasticity</code>	1.5
I_b	toxic baseline level	<code>clients.info_base_intensity</code>	0.60
I	toxic slope scale	<code>clients.info_intensity</code>	1.4
α	adversariality	<code>clients.alpha</code>	0.15
f	performative feedback gain	<code>clients.toxicity_feedback</code>	0.22
c_t	toxic spread-decay	<code>clients.info_spread_decay</code>	1.5
σ_s	informed signal noise	<code>clients.info_signal_noise</code>	0.6
w	quoting-cost convexity	<code>reward.quote_anchor_weight</code>	0.25
	quoting-cost anchor	<code>reward.quote_anchor_ref</code>	1.0
P	global P&L scale ($= \beta$)	<code>reward.pnl_scale</code>	1.0
λ_q	inventory-risk curvature	<code>reward.inv_risk_weight</code>	0.05
ρ	liquidity ratio at reference	state (<code>liquidity/liq_mean</code>)	≈ 1
$\bar{g}, \psi$	gate mean / adverse severity	derived (Gaussian integrals, §3.5)	-

8 Validation protocol (predict then verify)

1. **Predict.** For each f in `sweep_feedback.yaml`, solve (7) for h^* , then evaluate $m_{\text{pred}}(f) = \varepsilon\beta/\gamma$ from (3), (4), (5).
2. **Measure.** Run `measure_response_modulus` at h^* for the common-random-number BR-slope estimate $m_{\text{meas}}(f)$.
3. **Compare.** Report m_{pred} vs m_{meas} across the sweep with cross-seed IQR bands. The claim: the predicted boundary $(\alpha f)^*$ coincides with the measured crossing $m_{\text{meas}} = 1$, and $m_{\text{pred}} \approx m_{\text{meas}}$ in the contracting regime, diverging only in the saturated regime (Corollary 3).

A reference implementation belongs in `endo_market_v2/endo_market/analysis/analytic_boundary.py` (a pure function of `Config`), reusing `response_modulus.py` for step 2.

9 Honest caveats

- $\beta = \text{pnl_scale}$ is exact only under the frozen-environment FOC (A3). A future operator modelling $dD/d\phi$ (PerfGD, Priority 1.2) makes $h\tau(h)$ h -responsive and β gains a $-hc_t$ correction; the boundary form survives but β is no longer a bare constant. This is the bridge to 1.2, not a defect.
- $\lambda_q, \bar{g}, \psi$ are state-dependent. The boundary is stated at the probe reference state; the phase diagram (Priority 1.5) reports it as a band over the reference-state distribution.
- The cap (A4) is assumed slack. Deep in the unstable regime the cap re-convexifies the problem; the analytic modulus is valid up to and slightly past the boundary, not arbitrarily far into divergence.
- Single bond, zero skew (A1). The scalar boundary is the base case; the factor-model lift (1.5) and the multi-dealer $\varepsilon < \gamma/(N\beta)$ (1.3) build on it but are out of scope here.

References

- J. Perdomo, T. Zrnic, C. Mendler-Dünner, M. Hardt. *Performative Prediction*. ICML 2020. - the $m = \varepsilon\beta/\gamma$ criterion this work makes structural.
- M. Avellaneda, S. Stoikov. *High-frequency trading in a limit-order book*. Quant. Finance 2008; O. Guéant, C.-A. Lehalle, J. Fernández-Tapia (GLFT), 2013. - the exponential fill intensity $\lambda(\delta) = Ae^{-k\delta}$ underlying γ .
- See `../references.bib` and `literature/literature-raghav/README.md` for the full citation map (Barzykin et al. 2025 for the $d\tau/dh$ grounding; Cont–Kukanov–Stoikov for the informed-flow slope used in the triangulation extension of 1.1).